Tidy summaries

Every name of a package links to its help page, packages attach where the code loads them, and operators become "ligatures" that still copy as the original text:

```
library(dplyr)
library(ggplot2)

by_cyl <- function(data) {
  data |>
    group_by(cyl) |>
    summarise(mpg = mean(mpg), n = n())
}
ggplot(by_cyl(mtcars), aes(cyl, mpg)) + geom_col()
```

visualize

Inline, too: `stats::filter(1:10, rep(1/3, 3))`. R knows the answer: the cars average 20.1 miles per gallon. And a session:

```
> x <- c(2, 3, 5, 7, 11)
> sum(x[x > 2])
[1] 26
```

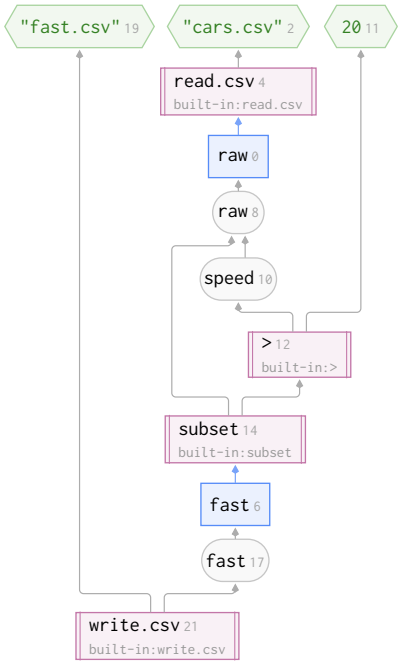
What flowR sees

Every name links to the definition it reads, the line that writes a file is marked, and flowR draws the dataflow of the listing:

```
raw <- read.csv("cars.csv")
fast <- subset(raw, speed > 20)
write.csv(fast, "fast.csv")
```

read

write


"fast.csv" 19
"cars.csv" 2
20 11
read.csv 4
built-in:read.csv
raw 0
raw 8
speed 10
> 12
built-in:>
subset 14
built-in:subset
fast 6
fast 17
write.csv 21
built-in:write.csv